Do Now · retrieval

Compilers, interpreters and assemblers

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What does LMC LDA do?

Q2

What is an executable?

Q3

What is a mnemonic?

Q4

Sorted [2,5,8,11,14,17,20,23,26], target23. Which midpoint indexes?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Understand — A compiler translates source code into another representation, often creating executable machine code via linking.

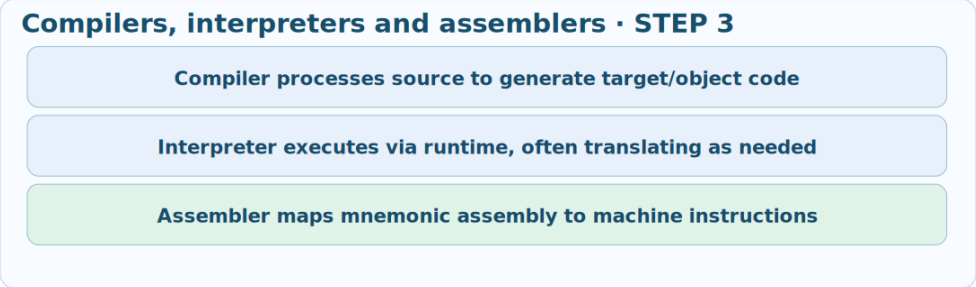
Core idea 2

Apply — An interpreter executes program statements with a runtime, often without a separate standalone compiled executable.

Core idea 3

Evaluate — An assembler translates assembly mnemonics and operands into processor machine instructions; translation strategies can overlap in modern environments.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

compiler — Translates high-level source into target code.

interpreter — Runs a program through an execution runtime.

assembler — Translates assembly to machine encoding.

object code — Translator-generated machine-level representation.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

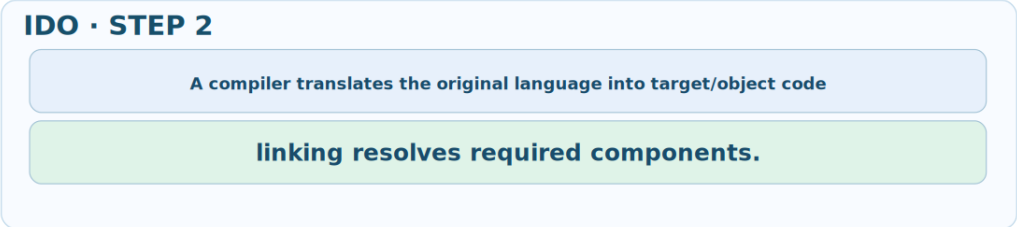
Problem

A developer wants to distribute a compiled native executable. Which translator and what does it produce?

Follow the worked method

1. Predict first — A compiler translates the original language into target/object code
2. Model answer — linking resolves required components.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict first

An assembler converts symbolic instruction representations into machine instruction encodings.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Identify compiler, interpreter and assembler by typical role. [3]

Question 2

Explain two differences between compiling native code and interpreting source at runtime. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Evaluate using an interpreter in a classroom programming exercise versus distributing a native compiled application. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

An assembler translates assembly-language mnemonics into machine instructions.

Use key vocabulary from memory

1. Explain compiler in your own words.

2. Explain interpreter in your own words.

3. Explain assembler in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.